

Day 29

T Flip-Flop (Edge-Triggered with Reset) in Verilog

Circuit Diagram :

T Flip-flop

Symbol

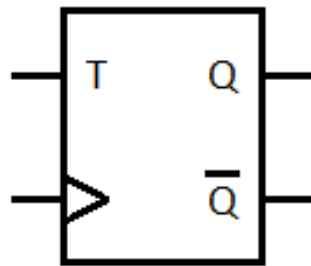
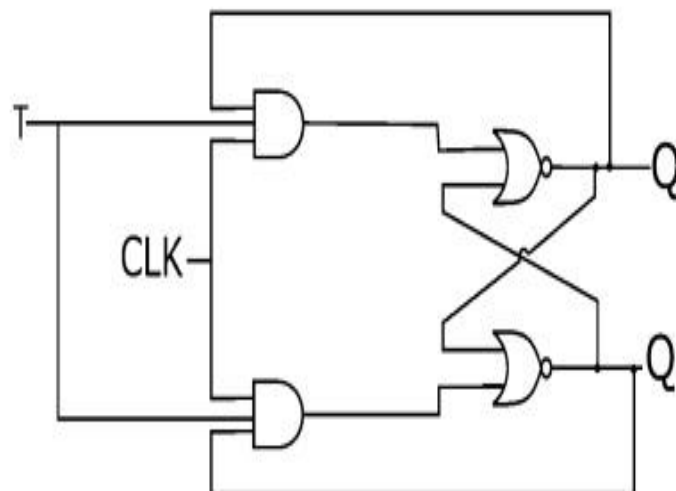


Table of truth:

T	Q	$\bar{Q}$
0	Q	$\bar{Q}$
1	$\bar{Q}$	Q
0	$\bar{Q}$	Q
1	Q	$\bar{Q}$



Verilog Code

T-FlipFlop verilog code

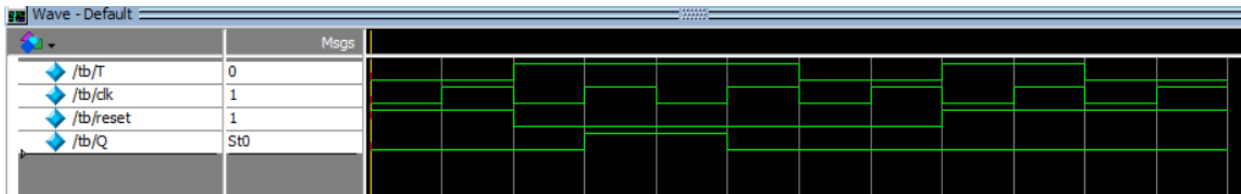
```
1 module T_FF(T,clk,reset,Q);
2   input T,
3   clk,reset;
4   output reg Q;
5   always @(posedge clk or posedge reset) begin
6     if(reset)
7       Q<=0;
8     else if(T)
9       Q<=~Q;
10  end
11 endmodule
```

Testbench Code

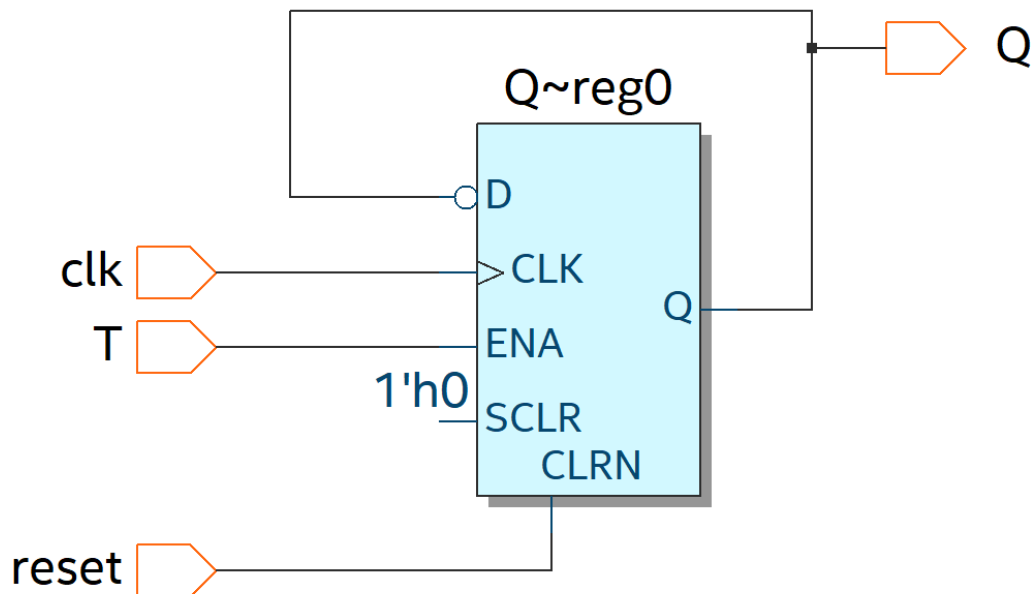
T-FlipFlop Tb code

```
1 module tb;
2   reg T,clk,reset;
3   wire Q;
4   T_FF dut(T,clk,reset,Q);
5   initial begin
6     clk=0;
7     forever #5 clk=~clk;
8   end
9   initial
10  begin
11    reset=1;T=0; #10
12    reset=0;T=1; #10
13    T=1; #10
14    reset=0;T=0; #10
15    reset=1;T=1; #10
16    T=0; #10
17    $finish;
18  end
19 endmodule
20
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim